

(1) $n = 1$ විට,

$$L.H.S \rightarrow \sum_{r=1}^n (r-1)(r+2) = 0$$

$$R.H.S \rightarrow \frac{1(1-1)(1+2)}{3} = 0$$

$$L.H.S = R.H.S$$

$n = p$ විට ප්‍රතික්ෂේපණය කරන බව

පෙන්වන්න.

$$\sum_{r=1}^p (r-1)(r+2) = \frac{p(p-1)(p+2)}{3}$$

$n = p+1$ විට,

$$\sum_{r=1}^{p+1} = \sum_{r=1}^p + \sum_{r=p+1}^{p+1}$$

$$= \frac{p(p-1)(p+2)}{3} + (p+1-1)(p+1+2)$$

$$= \frac{p(p^2 + 3p - 2) + 3p(p+3)}{3}$$

$$= \frac{p}{3} [p^2 + 3p - 2 + 3p + 9]$$

$$= \frac{p}{3} [p^2 + 6p + 7]$$

$$= \frac{p}{3} (p+1)(p+5)$$

$$= \frac{(p+1-1)(p+1)(p+1+2)}{3}$$

$n = 1$ විට ප්‍රතික්ෂේපණය කරන බව පෙන්වන්න.

$n = p$ විට ප්‍රතික්ෂේපණය කරන බව පෙන්වන්න.

$n = p+1$ විට ප්‍රතික්ෂේපණය කරන බව පෙන්වන්න.

එබැවින් අනුප්‍රාප්ත වූ ප්‍රතික්ෂේපණය අනුප්‍රාප්ත වූ $n \in \mathbb{Z}^+$ සඳහා

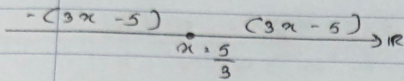
$$\sum_{r=1}^n (r-1)(r+2) = \frac{n(n-1)(n+2)}{3} \text{ බව, ප්‍රතික්ෂේපණය කරන බව පෙන්වන්න.}$$

(2)

$$|3x - 5| = y$$

$$3x - 5 = 0$$

$$x = \frac{5}{3}$$



$$x \leq \frac{5}{3} \text{ ඉටු,}$$

$$x > \frac{5}{3} \text{ ඉටු,}$$

$$y = 5 - 3x$$

$$y = 3x - 5$$

$$x = \frac{5}{3} \text{ ඉටු,}$$

$$x = 2 \text{ ඉටු,}$$

$$y = 0$$

$$y = 1$$

$$(5/3, 0)$$

$$(2, 1)$$

$$x = 0 \text{ ඉටු,}$$

$$y = 5 \quad (0, 5)$$

$$y = 2x + 3$$

$$0 = (1+1)(1-1) \quad 1+1=2$$

$$y = 0 \text{ ඉටු,}$$

$$x = -3/2$$

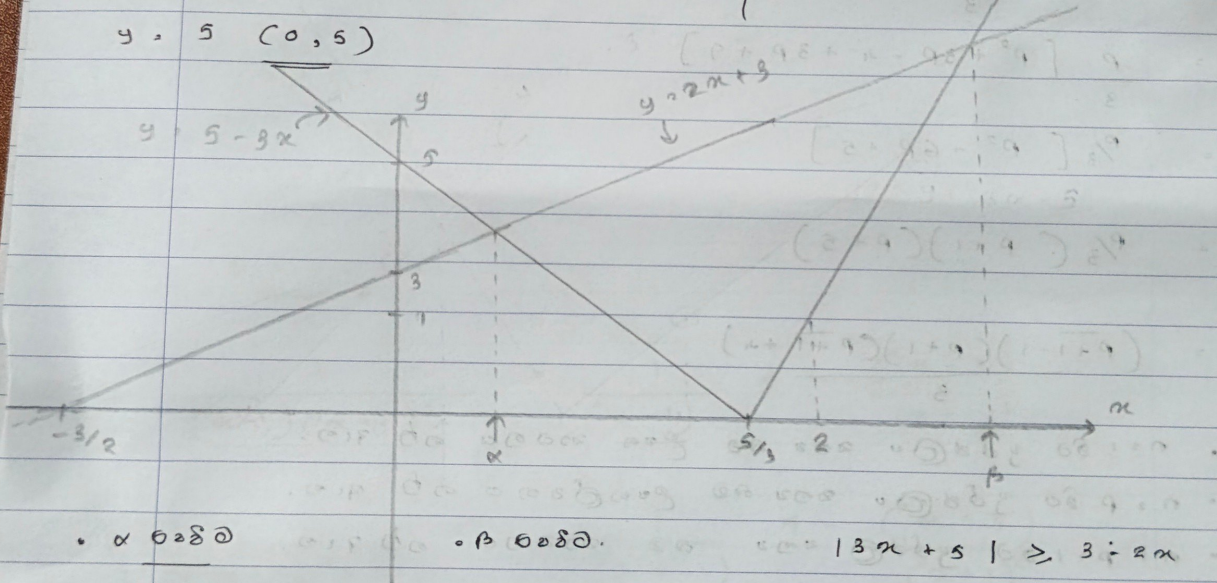
$$(0, -3/2, 0)$$

$$x = 0 \text{ ඉටු,}$$

$$y = 3$$

$$(0, 3)$$

$$y = 3x - 5$$



$$\alpha \text{ ඉටු}$$

$$\beta \text{ ඉටු}$$

$$|3x + 5| \geq 3 + 2x$$

$$\frac{2}{5} \leq x \leq 8$$

$$y = 5 - 3x \quad (1)$$

$$y = 3x - 5 \quad (1)$$

$$y = 2x + 3 \quad (2)$$

$$y = 2x + 3 \quad (2)$$

$$(1) = (2)$$

$$(1) = (2)$$

$$2x + 3 = 5 - 3x$$

$$3x - 5 = 2x + 3$$

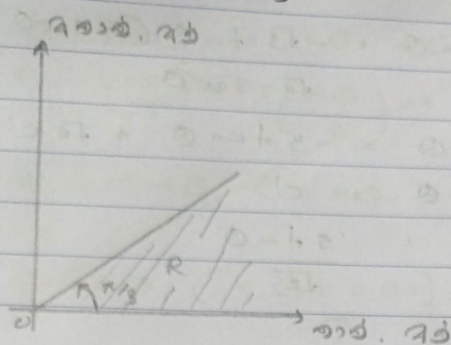
$$5x = 2$$

$$x = 8$$

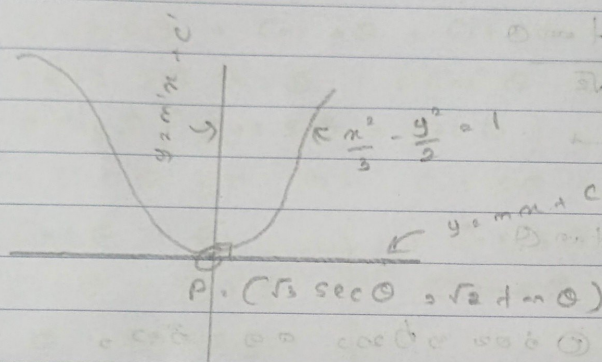
$$x = \frac{2}{5}$$

3

$$0 \leq \text{Arg} \leq \pi/3$$



6



$$x = \sqrt{3} \sec \theta$$

$$\frac{dx}{d\theta} = \sqrt{3} \sec \theta \tan \theta \quad \text{--- (1)}$$

$$y = \sqrt{2} \tan \theta$$

$$\frac{dy}{d\theta} = \sqrt{2} \sec^2 \theta \quad \text{--- (2)}$$

$$\frac{dy}{dx} = \frac{(2)}{(1)}$$

$$\frac{dy}{dx} = \frac{\sqrt{2} \sec^2 \theta}{\sqrt{3} \sec \theta \tan \theta}$$

$$\frac{dy}{dx} = \frac{\sqrt{2} \sec \theta}{\sqrt{3} \tan \theta} = m$$

$$y = mx + c \quad \leftarrow \text{Slope of tangent line}$$

$$\sqrt{2} \tan \theta = \frac{\sqrt{2} \sec \theta (\sqrt{3} \sec \theta)}{\sqrt{3} \tan \theta} + c$$

$$\sqrt{2} \tan \theta - \frac{\sqrt{2} \sec^2 \theta}{\tan \theta} = c$$

$$\sqrt{2} \tan^2 \theta - \sqrt{2} \sec^2 \theta = c \tan \theta$$

$$\sqrt{2} (\tan^2 \theta - (1 + \tan^2 \theta)) = c \tan \theta$$

$$\sqrt{2} (-1) = c \tan \theta$$

$$c = \frac{-\sqrt{2}}{\tan \theta}$$

• Slope of tangent line

$$y = \frac{\sqrt{2} \sec \theta x}{\sqrt{3} \tan \theta} - \frac{\sqrt{2}}{\tan \theta}$$

$$\frac{y}{\sqrt{2}} = \frac{\sec \theta x}{\sqrt{3} \tan \theta} - \frac{1}{\tan \theta}$$

$$\frac{y \tan \theta}{\sqrt{3}} = \frac{x \sec \theta}{\sqrt{3}} - 1$$

$$\frac{x \sec \theta}{\sqrt{3}} - \frac{y \tan \theta}{\sqrt{3}} = 1 \quad \leftarrow \text{Slope of tangent line}$$

• අනිශ්චිත අවකාශයේ $y = m'x + c'$

$$mm' = -1$$

$$\frac{\sqrt{2} \sec \theta (m')}{\sqrt{3} \tan \theta} = -1$$

$$m' = \frac{-\sqrt{3} \tan \theta}{\sqrt{2} \sec \theta}$$

$$y = m'x + c'$$

$$y = \frac{-\sqrt{3} \tan \theta}{\sqrt{2} \sec \theta} x + \frac{5 \tan \theta}{\sqrt{2}}$$

$$\sqrt{2} y \sec \theta = -\sqrt{3} \tan \theta x + 5 \tan \theta$$

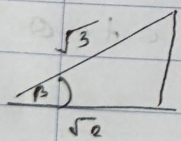
$$\sqrt{2} y = \frac{-\sqrt{3} \tan \theta x + 5 \tan \theta}{\sec \theta}$$

• අනිශ්චිත අවකාශයේ $(0, 5/2)$ ලක්ෂ්‍යයේ සිට ගමන් කරන රේඛාවේ $x = 0$ හි $y = 5/2$ අවකාශය.

$$\sqrt{2} \left(\frac{5}{2} \right) = \frac{-\sqrt{3} \tan \theta (0) + 5 \tan \theta}{\sec \theta}$$

$$\frac{5}{\sqrt{2}} = 5 \tan \theta$$

$$\tan \theta = \frac{1}{\sqrt{2}}$$



$$\theta = \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

$$\tan \theta = \tan \beta$$

$$\theta = n\pi + \beta$$

$$(n \in \mathbb{Z})$$

$$n = 0$$

$$\theta = \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

(10) $\cos 60^\circ + \cos 40^\circ + \cos 20^\circ + 1 = 0$
 $\cos 60^\circ + \cos 20^\circ + \cos 40^\circ + 1 = 0$
 $2 \cos \left(\frac{80^\circ}{2}\right) \cos \left(\frac{20^\circ}{2}\right) + \cos 40^\circ + 1 = 0$
 $2 \cos 40^\circ \cos 20^\circ + \cos 40^\circ + 1 = 0$
 $\cos 40^\circ (2 \cos 20^\circ + 1) + 1 = 0$

(10) $\cos 60^\circ + \cos 40^\circ + \cos 20^\circ + 1 = 0$
 $2 \cos 50^\circ \cos 10^\circ + 2 \cos^2 10^\circ - 1 + 1 = 0$
 $2 \cos 10^\circ [\cos 50^\circ + \cos 10^\circ] = 0$
 $2 \cos 10^\circ = 0$ or $\cos 50^\circ + \cos 10^\circ = 0$
 $\cos 10^\circ = 0$ $\left\{ \begin{array}{l} 2 \cos 30^\circ \cos 20^\circ = 0 \\ \cos 20^\circ = 0 \end{array} \right.$
 $\cos 10^\circ = \cos \frac{\pi}{2}$
 $10 = 2n\pi \pm \frac{\pi}{2}$
 $(n \in \mathbb{Z})$ $\left\{ \begin{array}{l} 2 \cos 30^\circ = 0 \text{ or } \cos 20^\circ = 0 \\ \cos 30^\circ = 0 \end{array} \right.$
 $\cos 30^\circ = 0$ $\cos 20^\circ = \cos \frac{\pi}{2}$
 $30 = 2m\pi \pm \frac{\pi}{2}$
 $10 = \frac{2m\pi \pm \pi}{3}$ $20 = 2n\pi \pm \frac{\pi}{2}$
 $(m \in \mathbb{Z})$ $(n \in \mathbb{Z})$

(11) a) $px^2 + qx + r = 0$

ଅବଲମ୍ବନୀୟ ଓ ଅବଲମ୍ବନୀୟ ଯଦି, ତେବେ ଯଦି ଯଦି

$\Delta \geq 0$ (1) (2) (3)

$b^2 - 4ac \geq 0$ (1) (2) (3)

$q^2 - 4pr \geq 0$ (1) (2) (3)

$$p^2 x^2 + p(3q - 2r)x + (2q - r)(q - r) + pr = 0 \quad (1)$$

$$D = b^2 - 4ac$$

$$D = p^2(3q - 2r)^2 - 4p^2[(2q - r)(q - r) + pr]$$

$$D = p^2[9q^2 + 4r^2 - 12qr - 4(2q^2 + r^2 - 3qr) + 4pr]$$

$$D = p^2[9q^2 + 4r^2 - 12qr - 8q^2 - 4r^2 + 12qr + 4pr]$$

$$D = p^2[q^2 + 4pr]$$

$$D = p^2[q^2 - 4pr]$$

$$q^2 - 4pr > 0 \text{ නම්}$$

$$D = p^2(q^2 - 4pr)$$

$$\downarrow$$

$$D = (+)$$

$$D > 0 \text{ වේ.}$$

∴ දෙකම තාත්වික 3 තීර්ණ වේ.

c) $f(x) = (x-1)(x-2)(x-3)$

$$R = \lambda(x-2)(x-3) + \mu(x-3)(x-1) + \gamma(x-1)(x-2)$$

බෙදීම ඇමුණාත්මක, ඔබාදම.

$$f(x) = g(x) \cdot \phi(x) + R$$

$$f(x) = (x-1)(x-2)(x-3) + R$$

$$\bullet \quad x-1 = 0$$

$$x = 1 \text{ තබා දීම}$$

$$f(x) = \lambda(-1)(-2) + \dots$$

$$f(1) = 2\lambda$$

$$\lambda = \frac{f(1)}{2}$$

$$\bullet \quad x-2 = 0$$

$$x = 2 \text{ තබා දීම}$$

$$f(2) = \mu(-1)(1)$$

$$\mu = -f(2)$$

$$\bullet \quad x-3 = 0$$

$$x = 3 \text{ තබා දීම}$$

$$f(3) = \gamma(2)(1)$$

$$\gamma = \frac{f(3)}{2}$$

$$f(x) = x^5 + kx^2 \text{ ඉහත බහුපදය}$$

$$f(1) = 2\lambda$$

$$\lambda = \frac{f(1)}{2}$$

$$m = -f(2) = -(x^5 + kx^2) = -(32 + 4k)$$

$$= \frac{x^5 + kx^2}{2} \quad f(3) = \frac{x^5 + kx^2}{2}$$

$$\gamma = \frac{f(3)}{2} = \frac{243 + 9k}{2}$$

$$R = \lambda(x-2)(x-3) + m(x-3)(x-1) + \gamma(x-1)(x-2)$$

x^2 හි සංගුණකය සමාන කරමු, $R = 0$ විය යුතුය.

$$\therefore \lambda(x^2 - 5x + 6) + m(x^2 - 4x + 3) + \gamma(x^2 - 3x + 2) = R$$

$$(\lambda + m + \gamma)x^2$$

x^2 හි සංගුණකය සමාන කරමු,

$$(\lambda + m + \gamma)x^2 = 0 \text{ විය යුතුය.}$$

$$\lambda + m + \gamma = 0$$

$$\frac{1+k}{2} + (-32 - 4k) + \frac{243 + 9k}{2} = 0$$

$$1+k - 64 - 8k + 243 + 9k = 0$$

$$2k + 180 = 0$$

$$2k = -180$$

$$k = -90$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1+k & -64-8k \\ 2k & 180 \end{bmatrix}$$

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(12) a) $U_{t+1} = \frac{2t-1}{t(t+1)(t+2)}$

$t(t+1)(t+2)$

$(t+3)U_{t+1} - tU_t = \frac{k}{(t+1)(t+2)}$

$(t+3)[2(t+1)-1] = \frac{(t)(2t-1)}{t(t+1)(t+2)} = \frac{k}{(t+1)(t+2)}$

$(t+3)[2(t+1)-1] = \frac{(t)(2t-1)}{t(t+1)(t+2)} = \frac{k}{(t+1)(t+2)}$

$2t+2-1 = (2t-1) = k$

$2t+1 = 2t+1 = k$

$k = 2$

(13) a) P^{-1} exists.

$\det P = y - \alpha$

$y > \alpha$

$\det P > 0$

$P^{-1} = \begin{pmatrix} y & -1 \\ -\alpha & 1 \end{pmatrix} \frac{1}{(y-\alpha)}$

$AP = \begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \alpha & y \end{bmatrix}$

$= \begin{bmatrix} 2\alpha - 1 & 2y - 1 \\ 2 + \alpha & 2 + y \end{bmatrix}$

$P^{-1}AP = \frac{1}{(y-\alpha)} \begin{bmatrix} y & -1 \\ -\alpha & 1 \end{bmatrix} \begin{bmatrix} 2\alpha - 1 & 2y - 1 \\ 2 + \alpha & 2 + y \end{bmatrix}$

$= \frac{2\alpha y - 9 + 2\alpha + 1}{(y-\alpha)}$

$= \begin{bmatrix} 2\alpha y - 9 + 2\alpha + 1 & 2y^2 - y - 4 - 9 \\ -2\alpha^2 + \alpha + 4 + \alpha & -2\alpha y + \alpha + 4 + y \end{bmatrix} = \begin{bmatrix} m & 0 \\ 0 & n \end{bmatrix}$

$(m, n \in \mathbb{Z})$

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$$2y^2 - 2y - 4 = 0 \quad \text{---} \quad -2x^2 + 2x + 4 = 0 \quad (*)$$

$$y^2 - y - 2 = 0 \quad \text{---} \quad x^2 - x - 2 = 0$$

$$y^2 + y - 2y - 2 = 0 \quad \text{---} \quad x^2 + x - 2x - 2 = 0$$

$$y(y+1) - 2(y+1) = 0 \quad \text{---} \quad x(x+1) - 2(x+1) = 0$$

$$(y+1)(y-2) = 0 \quad \text{---} \quad (x+1)(x-2) = 0$$

$$y = -1 \text{ or } y = 2 \text{ or } x = -1 \text{ or } x = 2$$

$$0 = (x+1)(x-2)$$

$$y > x \text{ or } y < x$$

$$y = 2 \text{ or } x = 2$$

$$x = -1 \text{ or } y = -1$$

$$A^2 = \begin{bmatrix} -1 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} = 9 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^4 = A^2(A^2) = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 81 & 0 \\ 0 & 81 \end{bmatrix} = 81 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b) \quad \left| \frac{z_1 - z_2}{z_1 + z_2} \right| = 1$$

$$|z_1 - z_2| = |z_1 + z_2|$$

$$|z_1 - z_2|^2 = |z_1 + z_2|^2$$

$$(z_1 - z_2)(\overline{z_1 - z_2}) = (z_1 + z_2)(\overline{z_1 + z_2})$$

$$(z_1 - z_2)(\overline{z_1} - \overline{z_2}) = (z_1 + z_2)(\overline{z_1} + \overline{z_2})$$

$$z_1\overline{z_1} + z_2\overline{z_2} - z_1\overline{z_2} - z_2\overline{z_1} = z_1\overline{z_1} + z_2\overline{z_2} + z_1\overline{z_2} + z_2\overline{z_1}$$

$$|z_1|^2 + |z_2|^2 = |z_1|^2 + |z_2|^2$$

(12) a) $2 - 2(\cos x - \sin x) - 2\sin 2x = 0$
 $2(1 - \cos x + \sin x) - \sin 2x = 0$

$$2 - 2\cos x + 2\sin x - 4\sin x \cos x = 0$$

$$1 - \cos x + \sin x - 2\sin x \cos x = 0$$

$$(1 - \cos x) + \sin x(1 - 2\cos x) = 0$$

$$(1 - \cos x)(1 + \sin x) = 0$$

$$\cos x = 1 \text{ or } \sin x = -1$$

$$\cos x = \cos 0^\circ \quad \sin x = -\sin \pi/2$$

$$x = 2n\pi \pm 0 \quad \sin x = \sin \pi/2$$

$$x = 2n\pi$$

$$(n \in \mathbb{Z})$$

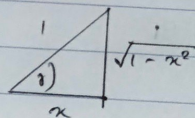
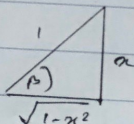
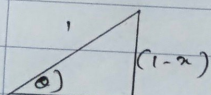
$$x = 2n\pi \pm \pi$$

$$x = m\pi + (-1)^m(\pi/2) \quad (m \in \mathbb{Z})$$

c) $\sin^{-1}(1-x) + \sin^{-1}(x) = \cos^{-1}(x)$

$\downarrow \quad \downarrow \quad \downarrow$
 $\theta \quad \beta \quad \gamma$

$\sin \theta = (1-x) \quad \sin \beta = x \quad \cos \gamma = x$



$$\theta + \beta = \gamma$$

$$\theta = \gamma - \beta$$

$$\sin(\gamma - \beta) = \sin \theta$$

$$\sin \gamma \cos \beta - \cos \gamma \sin \beta = \sin \theta$$

$$\sqrt{1-x^2} \left(\sqrt{1-x^2} \right) - x(x) = \sin \theta$$

$$1 - x^2 - x^2 = \sin \theta$$

$$1 - 2x^2 = \sin \theta$$

$$1 - 2x^2 = 1 - x$$

$$2x^2 - x = 0$$

$$x^2(2x - 1) = 0$$

$$x = 0 \text{ or } x = \frac{1}{2}$$

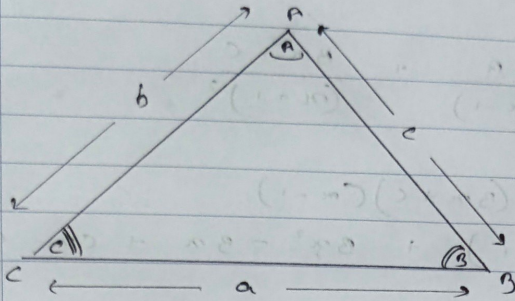
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$$x = 0 \text{ or } 0.5$$

$$\therefore x = \frac{1}{2}$$

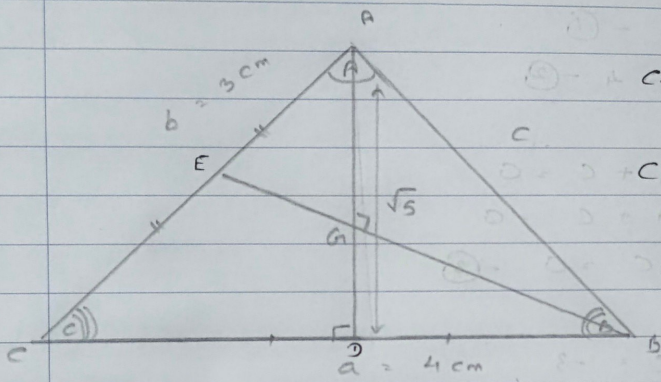
b)



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$= \frac{16 + 9 - 25}{2 \cdot 4 \cdot 3}$$

$$\cos C = \frac{0}{24} = 0$$

$$C = \cos^{-1}(0) = 90^\circ$$

AD ⊥ BC

$$AC^2 = CD^2 + AD^2$$

$$9 = 21 + AD^2$$

$$AD^2 = \sqrt{5}$$

AD ⊥ BC

$$AB^2 = AD^2 + BD^2$$

$$25 = 5 + 4$$

$$C = 3 \text{ cm}$$

$$\textcircled{1}, \cos C = \frac{25 - 9}{24}$$

$$\cos C = \frac{16}{24}$$

$$\cos C = \frac{4}{6} = \frac{2}{3}$$

$$C = \cos^{-1}\left(\frac{2}{3}\right)$$

15) a)
$$\frac{1}{(x-1)(x+1)^2} = \frac{A}{(x-1)} + \frac{Bx+C}{(x+1)^2}$$

$$1 = A(x+1)^2 + (Bx+C)(x-1)$$

$$1 = A(x^2 + 2x + 1) + Bx^2 - Bx + Cx - C$$

$$x^2; \rightarrow A + B = 0$$

$$A = -B \quad \text{--- (1)}$$

$$x^0; \rightarrow A - C = 1 \quad \text{--- (2)}$$

$$x^1; \rightarrow 2A - B + C = 0$$

$$2A + A + C = 0$$

$$3A + C = 0 \quad \text{--- (3)}$$

$$\text{(2) + (3)}$$

$$2A = 1 \quad C = -3$$

$$A = 1$$

$$B = -1$$

$$\frac{1}{(x-1)(x+1)^2} = \frac{1}{(x-1)} + \frac{(-x-3)}{(x+1)^2}$$

$$\frac{1}{(x-1)(x+1)^2} = \frac{1}{(x-1)} - \frac{(x+3)}{(x+1)^2}$$